THE GENERAL SINUS: TRIGONOMETRIC FUNCTIONS OF AN OBLIQUE PROJECTION

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1. Introduction

The classical sine and cosine of an angle α come from one construction. Take a point Z on the second side of the angle, project it straight down onto the first side, and read the sides of the right triangle that results [8]. The right angle is part of the construction, so familiar that it is rarely stated.

This paper changes one thing. The point Z is projected onto the first side along a line that makes a chosen angle ϕ with ZO, the *projection angle* (Figure 1). The triangle $OZ\overline{Z}$ is then oblique, with the angles α at O, ϕ at Z and $\pi-(\alpha+\phi)$ at the foot $\overline{Z}$. Its sides, divided by $[OZ]$, define the functions $\sin(\alpha,\phi)$ and $\cos(\alpha,\phi)$, and their quotient $\tan(\alpha,\phi)$. The law of sines [1] gives them at once,

$$\sin(\alpha,\phi)=\frac{\sin\alpha}{\sin(\alpha+\phi)},\qquad \cos(\alpha,\phi)=\frac{\sin\phi}{\sin(\alpha+\phi)},\qquad \tan(\alpha,\phi)=\frac{\sin\alpha}{\sin\phi},$$

and the classical functions return when $\alpha+\phi=\pi/2$.

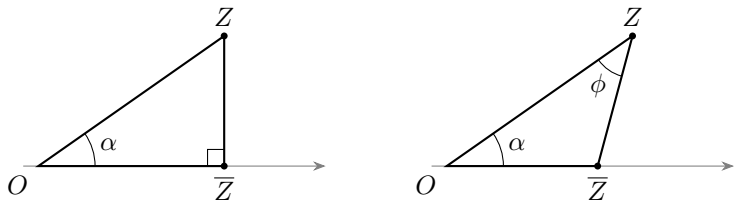
These formulas are elementary, and we do not claim otherwise. What makes them worth a systematic study is that they have a natural reading. When $[OZ]=1$, the projection line is parallel to a second axis through O at the angle $\overline{\phi}=\alpha+\phi$ to the first, and $\cos(\alpha,\phi)$ and $\sin(\alpha,\phi)$ are the coordinates of the point Z of the unit circle in the oblique frame of the two axes (Proposition 3.1). The classical functions are the coordinates of the unit circle in an orthonormal frame; the general functions are its coordinates in an oblique one. Oblique coordinates are as old as analytic geometry, but we are not aware of a study of the coordinate functions of the circle in such a frame as functions of two angles in their own right. The closest work concerns the generalized complex numbers, whose unit satisfies $i^2=p+qi$ [5, 6, 9]. In the elliptic case these algebras are the plane named in an oblique frame and the general functions are the coordinates of their unit circle; the arithmetic of numbers written in this form is developed in [3]. The name *generalized trigonometric functions* is also used for the eigenfunctions of the one-dimensional p-Laplacian [7]; those functions are unrelated to ours.

The general sine was introduced in [2]. The present paper gives a self-contained account, organised around the coordinate reading. Its main results are the following.

- In an oblique frame, the Pythagorean identity acquires one cross term, $\cos(\alpha,\phi)^2+\sin(\alpha,\phi)^2+2\cos(\alpha,\phi)\sin(\alpha,\phi)\cos(\alpha+\phi)=1$ (Proposition 3.2).
- The addition formulas come in two kinds: with the projection angle fixed, so that the frame turns with the studied angle (Section 5), and with the frame fixed (Proposition 6.2). All of the first kind reduce to a single identity of Ptolemy; the second kind follows from Euler's formula written in the oblique frame and includes De Moivre's formula.

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(a) straight down (b) at the angle ϕ

FIGURE 1. Two projections of a point Z onto the first side of the angle α. (a) Straight down: a right triangle, whose sides give $\cos\alpha$ and $\sin\alpha$. (b) Along a line making the angle ϕ with ZO: an oblique triangle with the angles α, ϕ and $\pi-(\alpha+\phi)$, whose sides give $\cos(\alpha,\phi)$ and $\sin(\alpha,\phi)$. Here $\alpha=35°$ and $\phi=40°$.

- Differentiation with the frame held fixed acts on the general sine and cosine as a quarter turn, $D^n f(x,y)=f(x+n\pi/2,\,y-n\pi/2)$, so that $D^2 f=-f$ as for the classical functions (Theorem 6.4).
- Each general function is a product of N copies of itself at scaled arguments, for every $N\geq 1$ (Theorem 7.1). Truncating each factor to second order gives rational functions which are again quotients of sines, in closed form, and which converge with an error of exact order N^{-2} (Theorem 7.3 and Corollary 7.4).

Section 2 defines the functions and their domains, Section 3 gives the coordinate reading, and Section 4 collects symmetries and special values. The results above occupy Sections 5 to 7. Every statement is checked symbolically or at 50 significant digits by an accompanying script (Section 8).

Notation. The studied angle is α, or x in analytic statements; the projection angle is ϕ, or y. Segment lengths are written $[PQ]$. The classical functions keep their one-variable notation, so that $\sin\alpha$ and $\sin(\alpha,\phi)$ are never confused.

2. The general functions

Let α be an angle with vertex O, let $Z\neq O$ be a point on its second side, and let $0<\phi<\pi-\alpha$. The line through Z that makes the angle ϕ with ZO meets the first side at a point $\overline{Z}$, the *projection of Z at the angle ϕ*. The triangle $OZ\overline{Z}$ has the angles α, ϕ and $\pi-(\alpha+\phi)$. All the triangles obtained in this way from points Z on the same side are similar, so the ratios of their sides depend on α and ϕ only.

Definition 2.1 (General trigonometric functions)**.** The *general sine, cosine and tangent* are the ratios

$$\sin(\alpha,\phi)=\frac{[Z\overline{Z}]}{[OZ]},\qquad \cos(\alpha,\phi)=\frac{[O\overline{Z}]}{[OZ]},\qquad \tan(\alpha,\phi)=\frac{[Z\overline{Z}]}{[O\overline{Z}]},$$

and csc, sec, cot are their reciprocals.

Theorem 2.2 (Analytic form)**.** *For $0<\alpha$, $0<\phi$ and $\alpha+\phi<\pi$,*

$$\sin(\alpha,\phi)=\frac{\sin\alpha}{\sin(\alpha+\phi)},\qquad \cos(\alpha,\phi)=\frac{\sin\phi}{\sin(\alpha+\phi)},\qquad \tan(\alpha,\phi)=\frac{\sin\alpha}{\sin\phi}. \tag{2.1}$$

Proof. By the law of sines in $OZ\overline{Z}$, $[Z\overline{Z}]/\sin\alpha=[O\overline{Z}]/\sin\phi=[OZ]/\sin(\pi-\alpha-\phi)$, and $\sin(\pi-\alpha-\phi)=\sin(\alpha+\phi)$. □

From here on, (2.1) *defines* the general functions for all real α and ϕ at which the denominators do not vanish: $\sin(\alpha,\phi)$ and $\cos(\alpha,\phi)$ need $\alpha+\phi\notin\pi\mathbb{Z}$, and $\tan(\alpha,\phi)$ needs

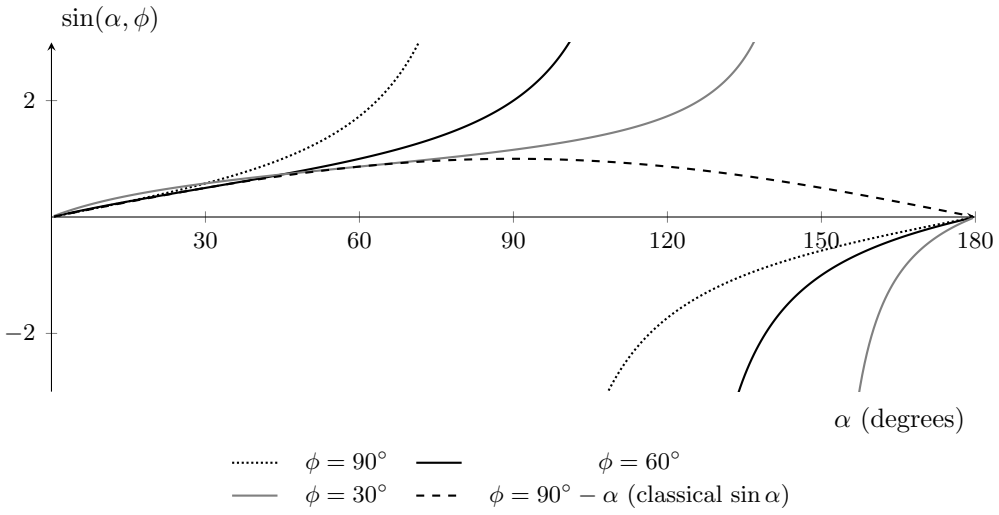


FIGURE 2. The general sine as a function of α for three fixed projection angles. It has a pole at $\alpha = 180^\circ - \phi$, where the projection line becomes parallel to the first side. Along the line $\alpha + \phi = 90^\circ$ (dashed) it is the classical sine. For $\phi = 90^\circ$ it is $\tan\alpha$ (dotted), by Proposition 4.3.

$\phi \notin \pi\mathbb{Z}$. The reciprocals are

$$\csc(\alpha,\phi) = \frac{\sin(\alpha+\phi)}{\sin\alpha}, \qquad \sec(\alpha,\phi) = \frac{\sin(\alpha+\phi)}{\sin\phi}, \qquad \cot(\alpha,\phi) = \frac{\sin\phi}{\sin\alpha},$$

defined where their denominators do not vanish (for csc and sec we also keep $\alpha+\phi \notin \pi\mathbb{Z}$, so that they are reciprocals of finite non-zero values). For α and ϕ outside the triangle range, the general functions keep the geometric meaning of Proposition 3.1, as signed coordinates.

Corollary 2.3 (Classical case)**.** *If $\alpha + \phi = \pi/2$, then*

$$\sin(\alpha,\phi) = \sin\alpha, \qquad \cos(\alpha,\phi) = \cos\alpha, \qquad \tan(\alpha,\phi) = \tan\alpha.$$

Proof. Then $\sin(\alpha+\phi) = 1$ and $\sin\phi = \cos\alpha$. □

The condition $\alpha + \phi = \pi/2$ says that the triangle has its right angle at $\overline{Z}$. The general functions therefore form a one-parameter family through the classical ones, the parameter recording the direction of projection.

Example. For $\alpha = \pi/4$ and $\phi = \pi/3$,

$$\sin\Big(\frac{\pi}{4},\frac{\pi}{3}\Big) = \frac{\sin(\pi/4)}{\sin(7\pi/12)} = \frac{\sqrt{2}/2}{(\sqrt{6}+\sqrt{2})/4} = \sqrt{3}-1 \approx 0.732,$$

$$\cos\Big(\frac{\pi}{4},\frac{\pi}{3}\Big) = \frac{\sqrt{3}/2}{(\sqrt{6}+\sqrt{2})/4} = \frac{3\sqrt{2}-\sqrt{6}}{2} \approx 0.897.$$

3. The coordinate reading

Draw the unit circle about O together with a second axis through O, the *composer axis* Com, making the angle $\overline{\phi}$ with the x-axis, $0 < \overline{\phi} < \pi$ (Figure 3). We call this configuration the *composed circle*. For the point Z of the circle at the angle α, projecting Z onto the x-axis parallel to the composer axis is the projection at the angle $\phi = \overline{\phi} - \alpha$, since the projection line and ZO meet at that angle. The three angles are therefore tied by

$$\overline{\phi} = \alpha + \phi. \tag{3.1}$$

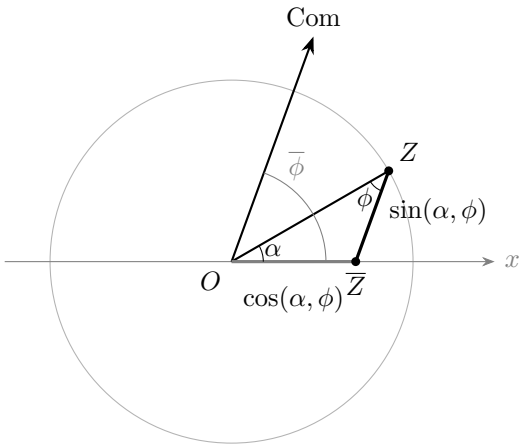


FIGURE 3. The composed circle. The point Z of the unit circle at the angle α, projected onto the x-axis parallel to the composer axis ($\overline{\phi} = \alpha + \phi$), has the coordinate $\cos(\alpha, \phi)$ along the x-axis and $\sin(\alpha, \phi)$ along the composer axis (Proposition 3.1). Here $\alpha = 30°$ and $\overline{\phi} = 70°$.

Proposition 3.1 (Coordinates in an oblique frame)**.** *Let $0 < \overline{\phi} < \pi$, let $\vec{u} = (\cos\overline{\phi}, \sin\overline{\phi})$ be the unit vector of the composer axis, and let $Z = (\cos\alpha, \sin\alpha)$, with $\phi = \overline{\phi} - \alpha$. Then*

$$\overrightarrow{OZ} = \cos(\alpha, \phi)\, \vec{e}_1 + \sin(\alpha, \phi)\, \vec{u},$$

where $\vec{e}_1 = (1, 0)$. That is, $\cos(\alpha, \phi)$ and $\sin(\alpha, \phi)$ are the coordinates of Z in the frame $(\vec{e}_1, \vec{u})$, for every real α.

Proof. Solve $(\cos\alpha, \sin\alpha) = a(1, 0) + b(\cos\overline{\phi}, \sin\overline{\phi})$. The second components give $b = \sin\alpha / \sin\overline{\phi}$, and the first give $a = \cos\alpha - b\cos\overline{\phi} = \sin(\overline{\phi} - \alpha)/\sin\overline{\phi}$. With $\overline{\phi} = \alpha + \phi$ these are $\sin(\alpha, \phi)$ and $\cos(\alpha, \phi)$. Since $\sin\overline{\phi} \neq 0$, the solution is unique. □

The proposition gives the general functions a meaning for all real α, as signed coordinates, and it explains their poles: as $\overline{\phi} \to 0$ or π the composer axis falls onto the x-axis and the oblique frame degenerates. The length of $\overrightarrow{OZ}$ written in the frame gives the oblique form of Pythagoras.

Proposition 3.2 (Pythagorean identity of an oblique frame)**.** *Whenever $\alpha + \phi \notin \pi\mathbb{Z}$,*

$$\cos(\alpha, \phi)^2 + \sin(\alpha, \phi)^2 + 2\cos(\alpha, \phi)\sin(\alpha, \phi)\cos(\alpha + \phi) = 1. \tag{3.2}$$

Proof. $|a\vec{e}_1 + b\vec{u}|^2 = a^2 + b^2 + 2ab\cos\overline{\phi}$ and $|\overrightarrow{OZ}| = 1$. Equivalently, (3.2) is the law of cosines at $\overline{Z}$ in the triangle $OZ\overline{Z}$, whose angle there is $\pi - \overline{\phi}$. □

For $\overline{\phi} = \pi/2$ the cross term vanishes and (3.2) is $\cos^2\alpha + \sin^2\alpha = 1$. In general the point $(\cos(\alpha, \phi), \sin(\alpha, \phi))$, plotted with its two coordinates at right angles, runs along an ellipse as α varies with $\overline{\phi}$ fixed; its axes lie on the diagonals [3].

4. Symmetries and special values

All statements in this section follow directly from (2.1); we indicate the one-line arguments.

Proposition 4.1 (Exchange and reciprocals)**.** *On the respective domains,*

$$\sin(\alpha, \phi) = \cos(\phi, \alpha), \qquad \tan(\alpha, \phi) = \cot(\phi, \alpha), \qquad \csc(\alpha, \phi) = \sec(\phi, \alpha),$$

and the reciprocals are general functions themselves,

$$\csc(\alpha,\phi)=\sin(\alpha+\phi,\,-\phi),\qquad \sec(\alpha,\phi)=\tan(\alpha+\phi,\,\phi),$$
$$\cot(\alpha,\phi)=\cos(\alpha-\phi,\,\phi).$$

Proof. Exchanging α and ϕ in (2.1) exchanges the numerators $\sin\alpha$ and $\sin\phi$. For the second line, $\sin(\alpha+\phi,-\phi)=\sin(\alpha+\phi)/\sin\alpha$, $\tan(\alpha+\phi,\phi)=\sin(\alpha+\phi)/\sin\phi$ and $\cos(\alpha-\phi,\phi)=\sin\phi/\sin\alpha$. □

Proposition 4.2 (Signs and shifts)**.** *On the respective domains, for all integers p and q,*

$$\sin(-\alpha,-\phi)=\sin(\alpha,\phi),\qquad \cos(-\alpha,-\phi)=\cos(\alpha,\phi),$$
$$\sin(-\alpha,\phi)=\sin(\alpha,-\phi),$$
$$\sin(\alpha+p\pi,\,\phi+q\pi)=(-1)^q\sin(\alpha,\phi),$$
$$\cos(\alpha+p\pi,\,\phi+q\pi)=(-1)^p\cos(\alpha,\phi),$$
$$\tan(\alpha+p\pi,\,\phi+q\pi)=(-1)^{p+q}\tan(\alpha,\phi).$$

Proof. The sine is odd, and $\sin(t+m\pi)=(-1)^m\sin t$. □

In particular the general functions are 2π-periodic in each variable. The shift rule is not symmetric in the two variables: a half turn of the projection angle changes the sign of the general sine, and a half turn of the studied angle that of the general cosine.

Proposition 4.3 (Special values)**.** *On the respective domains,*

$$\sin(\alpha,\tfrac{\pi}{2})=\tan\alpha,\qquad \cos(\alpha,\tfrac{\pi}{2})=\sec\alpha,\qquad \tan(\alpha,\tfrac{\pi}{2})=\sin\alpha,$$
$$\sin(\theta,\theta)=\cos(\theta,\theta)=\frac{1}{2\cos\theta},\qquad \tan(\theta,\theta)=1.$$

Proof. $\sin(\alpha+\pi/2)=\cos\alpha$, and $\sin 2\theta=2\sin\theta\cos\theta$. □

At $\theta=\pi/3$ all six general functions of (θ,θ) equal 1: the triangle is equilateral. At $\theta=\pi/5$, $\sin(\theta,\theta)=\cos(\theta,\theta)=1/(2\cos 36^\circ)=\tau^{-1}$, where $\tau=(1+\sqrt{5})/2$ is the golden ratio, and the triangle is the golden triangle with angles 36°, 36°, 108°.

5. Addition formulas

The addition formulas of the general functions all reduce, after clearing denominators, to one identity among ordinary sines.

Lemma 5.1 (Ptolemy's identity)**.** *For all real x, y and z,*

$$\sin(x+y)\sin(y+z)=\sin x\,\sin z+\sin y\,\sin(x+y+z). \tag{5.1}$$

Proof. By $2\sin P\sin Q=\cos(P-Q)-\cos(P+Q)$, both sides equal $\frac{1}{2}\big[\cos(x-z)-\cos(x+2y+z)\big]$. □

For a cyclic quadrilateral with consecutive arcs $2x$, $2y$, $2z$ and $2\pi-2(x+y+z)$, inscribed in a circle of diameter 1, the chords are $\sin x$, $\sin y$, $\sin z$, $\sin(x+y+z)$ and the diagonals $\sin(x+y)$, $\sin(y+z)$, and (5.1) is Ptolemy's theorem; hence the name.

Theorem 5.2 (Addition in the studied angle)**.** *On the domains where both sides are defined,*

$$\sin(\alpha+\beta,\phi)=\sin(\alpha,\phi)+\cos(\alpha,\phi)\,\sin(\beta,\phi+\alpha), \tag{5.2}$$
$$\cos(\alpha+\beta,\phi)=\cos(\alpha,\phi)\cos(\beta,\phi)+\sin(\alpha,\phi)\sin(\beta,\phi)\cos(\alpha+\beta,\phi), \tag{5.3}$$
$$\tan(\alpha+\beta,\phi)=\tan(\alpha,\phi)\cos\beta+\tan(\beta,\phi)\cos\alpha. \tag{5.4}$$

Proof. Multiply (5.2) by $\sin(\alpha+\phi)\sin(\alpha+\beta+\phi)$: it becomes $\sin(\alpha+\beta)\sin(\alpha+\phi) = \sin\alpha\,\sin(\alpha+\beta+\phi)+\sin\phi\,\sin\beta$, which is (5.1) with $(x,y,z)=(\beta,\alpha,\phi)$. Multiply (5.3) by $\sin(\alpha+\phi)\sin(\beta+\phi)\sin(\alpha+\beta+\phi)/\sin\phi$: it becomes $\sin(\alpha+\phi)\sin(\beta+\phi) = \sin\phi\,\sin(\alpha+\beta+\phi)+\sin\alpha\,\sin\beta$, which is (5.1) with $(x,y,z)=(\alpha,\phi,\beta)$. Finally (5.4) is $\sin(\alpha+\beta) = \sin\alpha\cos\beta+\cos\alpha\sin\beta$ divided by $\sin\phi$. □

The cosine formula (5.3) contains $\cos(\alpha+\beta,\phi)$ on both sides. Solving for it, and inverting, gives two closed forms:

$$\cos(\alpha+\beta,\phi) = \frac{\cos(\alpha,\phi)\cos(\beta,\phi)}{1-\sin(\alpha,\phi)\sin(\beta,\phi)}, \tag{5.5}$$

$$\sec(\alpha+\beta,\phi) = \sec(\alpha,\phi)\sec(\beta,\phi)-\tan(\alpha,\phi)\tan(\beta,\phi). \tag{5.6}$$

The second is the exact analogue of $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$, with the secant and tangent in the roles of the cosine and sine.

Theorem 5.3 (Addition in the projection angle)**.** *On the domains where both sides are defined,*

$$\sin(\alpha,\phi+\theta) = \sin(\alpha,\phi)\,\sin(\alpha+\phi,\theta), \tag{5.7}$$

$$\cos(\alpha,\phi+\theta) = \cos(\alpha,\phi)+\sin(\alpha,\phi)\,\cos(\alpha+\phi,\theta). \tag{5.8}$$

Proof. The right side of (5.7) is $\frac{\sin\alpha}{\sin(\alpha+\phi)}\cdot\frac{\sin(\alpha+\phi)}{\sin(\alpha+\phi+\theta)}$. Multiplying (5.8) by $\sin(\alpha+\phi)\sin(\alpha+\phi+\theta)$ gives $\sin(\alpha+\phi)\sin(\phi+\theta) = \sin\phi\,\sin(\alpha+\phi+\theta)+\sin\alpha\,\sin\theta$, which is (5.1) with $(x,y,z)=(\alpha,\phi,\theta)$. □

Formula (5.7) composes two oblique projections: projecting at the angle ϕ and then, from the new point, at the angle θ, multiplies the general sines, just as similar triangles multiply ratios. Formulas (5.2)–(5.4) have a geometric proof on the composed circle as well, by cutting the triangle of $\alpha+\beta$ into the triangles of α and β; the algebraic proof above is shorter and covers all signs at once.

Corollary 5.4 (Double and half angles)**.** *On the domains where both sides are defined,*

$$\sin(2\alpha,\phi) = \sin(\alpha,\phi)\big(1+\cos(2\alpha,\phi)\big), \qquad \cos(2\alpha,\phi) = \frac{\cos(\alpha,\phi)^2}{1-\sin(\alpha,\phi)^2},$$

$$\sec(2\alpha,\phi) = \sec(\alpha,\phi)^2-\tan(\alpha,\phi)^2, \qquad \tan(2\alpha,\phi) = 2\tan(\alpha,\phi)\cos\alpha,$$

and, replacing α by $\alpha/2$, $\cos(\alpha,\phi) = \cos(\frac{\alpha}{2},\phi)^2\big/\big(1-\sin(\frac{\alpha}{2},\phi)^2\big)$.

Proof. Put $\beta=\alpha$ in Theorem 5.2, (5.5) and (5.6). For the first formula,

$$\cos(\alpha,\phi)\sin(\alpha,\phi+\alpha) = \frac{\sin\phi}{\sin(\alpha+\phi)}\cdot\frac{\sin\alpha}{\sin(2\alpha+\phi)} = \sin(\alpha,\phi)\cos(2\alpha,\phi). \qquad \square$$

6. The fixed frame: multiplication and derivatives

In analytic statements we write (x,y) for (α,ϕ) and f_x, f_y for partial derivatives.

Proposition 6.1 (Partial derivatives)**.** *On the domains of the functions,*

$$\partial_x\sin(x,y) = \frac{\cos(x,y)}{\sin(x+y)}, \qquad \partial_y\sin(x,y) = -\sin(x,y)\cot(x+y),$$

$$\partial_x\cos(x,y) = -\cos(x,y)\cot(x+y), \qquad \partial_y\cos(x,y) = \frac{\sin(x,y)}{\sin(x+y)},$$

$$\partial_x\tan(x,y) = \frac{\cos x}{\sin y}, \qquad \partial_y\tan(x,y) = -\tan(x,y)\cot y,$$

and the second derivatives are

$$\begin{aligned}
\partial_x^2 \sin(x,y) &= -2\cot(x+y)\,\partial_x \sin(x,y),\\
\partial_y^2 \sin(x,y) &= \sin(x,y)\big(1+2\cot^2(x+y)\big),\\
\partial_x^2 \cos(x,y) &= \cos(x,y)\big(1+2\cot^2(x+y)\big),\\
\partial_y^2 \cos(x,y) &= -2\cot(x+y)\,\partial_y \cos(x,y),\\
\partial_x^2 \tan(x,y) &= -\tan(x,y), \qquad \partial_y^2 \tan(x,y) = \tan(x,y)\big(1+2\cot^2 y\big).
\end{aligned}$$

Proof. Direct differentiation of (2.1). For instance $\partial_x \sin(x,y) = \big(\cos x \sin(x+y) - \sin x \cos(x+y)\big)/\sin^2(x+y) = \sin y/\sin^2(x+y)$, and $\partial_y^2\big(\sin x/\sin(x+y)\big) = \sin x\big(\sin^2(x+y) + 2\cos^2(x+y)\big)/\sin^3(x+y)$. □

The partial derivatives move the frame: changing x with y fixed turns the composer axis, since $\overline{\phi} = x+y$. So do the addition formulas of Section 5, in which the projection angle ϕ is fixed and the frame angle $\overline{\phi} = \alpha + \phi$ changes with α. The rest of this section keeps the frame fixed and moves the point along the circle. For a fixed $\overline{\phi} \in (0,\pi)$ write

$$C_\theta = \cos(\theta,\, \overline{\phi} - \theta) = \frac{\sin(\overline{\phi} - \theta)}{\sin \overline{\phi}}, \qquad S_\theta = \sin(\theta,\, \overline{\phi} - \theta) = \frac{\sin\theta}{\sin\overline{\phi}}, \tag{6.1}$$

the coordinates of the point of the unit circle at the angle θ in the frame $(\vec{e}_1, \vec{u})$ (Proposition 3.1). Identify the plane with $\mathbb{C}$, so that $\vec{e}_1 = 1$ and $\vec{u} = u = e^{i\overline{\phi}}$. Then Proposition 3.1 reads

$$e^{i\theta} = C_\theta + S_\theta\, u, \tag{6.2}$$

which is Euler's formula written in the oblique frame. Since u is a root of $t^2 - 2\cos\overline{\phi}\, t + 1 = (t - e^{i\overline{\phi}})(t - e^{-i\overline{\phi}})$,

$$u^2 = 2\cos\overline{\phi}\; u - 1. \tag{6.3}$$

Proposition 6.2 (Addition in a fixed frame)**.** *For a fixed $\overline{\phi} \in (0,\pi)$ and all real α, β,*

$$C_{\alpha+\beta} = C_\alpha C_\beta - S_\alpha S_\beta, \tag{6.4}$$

$$S_{\alpha+\beta} = C_\alpha S_\beta + S_\alpha C_\beta + 2\cos\overline{\phi}\; S_\alpha S_\beta, \tag{6.5}$$

and for every integer n,

$$\big(C_\theta + S_\theta\, u\big)^n = C_{n\theta} + S_{n\theta}\, u. \tag{6.6}$$

Proof. Multiply $e^{i\alpha} = C_\alpha + S_\alpha u$ by $e^{i\beta} = C_\beta + S_\beta u$ and reduce u^2 with (6.3):

$$e^{i(\alpha+\beta)} = \big(C_\alpha C_\beta - S_\alpha S_\beta\big) + \big(C_\alpha S_\beta + S_\alpha C_\beta + 2\cos\overline{\phi}\, S_\alpha S_\beta\big)\, u.$$

Since 1 and u are linearly independent over $\mathbb{R}$, the coordinates of $e^{i(\alpha+\beta)}$ in the frame are unique, and they are $C_{\alpha+\beta}$ and $S_{\alpha+\beta}$ by (6.2). Formula (6.6) is $(e^{i\theta})^n = e^{in\theta}$ read in the same way. □

For $\overline{\phi} = \pi/2$, (6.4) and (6.5) are the classical addition formulas, and (6.6) is De Moivre's formula. In the oblique frame the sine formula acquires the cross term $2\cos\overline{\phi}\, S_\alpha S_\beta$, the same cross term as in the Pythagorean identity (3.2), which in this notation reads $C_\theta^2 + 2\cos\overline{\phi}\, C_\theta S_\theta + S_\theta^2 = 1$. With $\theta = \overline{\phi}$ in (6.6), $C_{n\overline{\phi}} = -\sin((n-1)\overline{\phi})/\sin\overline{\phi}$ and $S_{n\overline{\phi}} = \sin(n\overline{\phi})/\sin\overline{\phi} = U_{n-1}(\cos\overline{\phi})$, a Chebyshev polynomial of the second kind, so that the powers of u in its own frame are $u^n = -U_{n-2}(\cos\overline{\phi}) + U_{n-1}(\cos\overline{\phi})\, u$ for $n \geq 2$.

The derivative that moves the point with the frame held fixed is the derivative along the lines $x + y = \text{const}$.

Definition 6.3 (Derivative in a fixed frame)**.** For a differentiable function f of (x, y), let

$$Df(x,y) = \frac{\mathrm{d}}{\mathrm{d}t}\, f(x+t,\, y-t)\Big|_{t=0} = \partial_x f(x,y) - \partial_y f(x,y).$$

Along the path $t \mapsto (x+t, y-t)$ the studied angle grows while the projection angle shrinks by the same amount, so $\overline{\phi} = x+y$ is constant: the point Z turns on the unit circle and the oblique frame does not move. This is the operation that the classical derivative of $\sin\alpha$ performs on the unit circle with its orthonormal frame.

Theorem 6.4 (Quarter-turn shift)**.** *For every integer $n \geq 0$ and (x, y) in the domain,*

$$D^n \sin(x,y) = \sin\Big(x + \frac{n\pi}{2},\; y - \frac{n\pi}{2}\Big), \tag{6.7}$$

$$D^n \cos(x,y) = \cos\Big(x + \frac{n\pi}{2},\; y - \frac{n\pi}{2}\Big). \tag{6.8}$$

In particular $D^2 \sin(x,y) = -\sin(x,y)$ and $D^2 \cos(x,y) = -\cos(x,y)$.

Proof. Along the path, $\sin(x+t, y-t) = \sin(x+t)/\sin(x+y)$ with a constant denominator, so $D^n \sin(x,y) = \sin(x+n\pi/2)/\sin(x+y)$, which is the general sine at $(x+n\pi/2,\, y-n\pi/2)$ because the sum of the two arguments is still $x+y$. The same argument applies to $\cos(x,y) = \sin y/\sin(x+y)$, with $\frac{\mathrm{d}^n}{\mathrm{d}t^n}\sin(y-t) = \sin(y-t-n\pi/2)$. For $n=2$, Proposition 4.2 gives $\sin(x+\pi, y-\pi) = -\sin(x,y)$ and $\cos(x+\pi, y-\pi) = -\cos(x,y)$. □

Corollary 6.5 (Primitives in a fixed frame)**.** *For every integer $n \geq 1$, the functions $\sin(x - n\pi/2,\, y + n\pi/2)$ and $\cos(x - n\pi/2,\, y + n\pi/2)$ satisfy $D^n F = \sin(x,y)$ and $D^n F = \cos(x,y)$ respectively; every n-th primitive along the lines $x+y = \text{const}$ differs from them by a polynomial of degree less than n in t whose coefficients depend only on $x+y$.*

Proof. Apply Theorem 6.4 to the shifted point. Two n-th primitives along a line differ by a solution of $\mathrm{d}^n g/\mathrm{d}t^n = 0$. □

Theorem 6.4 says two things. With the frame held fixed, the general cosine and sine obey the same equation $D^2 f = -f$ as the classical pair. And one derivative is a quarter turn of the point: $D\sin(x,y) = \sin(x+\pi/2,\, y-\pi/2)$ is the general sine of the point a quarter turn further on the circle, read in the same frame. In the notation (6.1), this is the derivative of (6.2), $\frac{\mathrm{d}}{\mathrm{d}\theta}\big(C_\theta + S_\theta u\big) = i\, e^{i\theta} = C_{\theta+\pi/2} + S_{\theta+\pi/2}\, u$. The general tangent does not have this property, because both its numerator and its denominator move along the path. Writing $\tan(x+t, y-t) = \sin(x+y)\cot(y-t) - \cos(x+y)$ gives, for $n \geq 1$,

$$D\tan(x,y) = \frac{\sin(x+y)}{\sin^2 y}, \qquad D^n \tan(x,y) = \sin(x+y)\Big(-\frac{\mathrm{d}}{\mathrm{d}y}\Big)^{n-1} \csc^2 y.$$

7. Product identities and the General Sinus Transform

This section writes each general function as a finite product of copies of itself at scaled arguments. The identities are exact and hold for any number of factors. Replacing every factor by a ratio of two quadratic polynomials then gives a rational approximation, which we call the *General Sinus Transform*.

The identities rest on the classical product of sines [4, §1.39]: for every integer $N \geq 1$ and every real ϑ,

$$\prod_{k=0}^{N-1} \sin\Big(\vartheta + \frac{\pi k}{N}\Big) = \frac{\sin(N\vartheta)}{2^{N-1}}. \tag{7.1}$$

Theorem 7.1 (Product identities)**.** *Let $N \geq 1$ be an integer and let $x, y \in \mathbb{R}$. If $x+y \notin \pi\mathbb{Z}$, then*

$$\sin(x,y) = \prod_{k=0}^{N-1} \sin\Big(\frac{x+\pi k}{N},\, \frac{y}{N}\Big), \tag{7.2}$$

$$\cos(x,y) = \prod_{k=0}^{N-1} \cos\Big(\frac{x}{N},\, \frac{y+\pi k}{N}\Big), \tag{7.3}$$

and if $y \notin \pi\mathbb{Z}$, then

$$\tan(x,y) = \prod_{k=0}^{N-1} \tan\Big(\frac{x+\pi k}{N},\, \frac{y+\pi k}{N}\Big). \tag{7.4}$$

Every factor lies in the domain of its function.

Proof. Apply (7.1) with $N\vartheta = x$, $N\vartheta = y$ and $N\vartheta = x+y$. The factors 2^{N-1} cancel in each quotient:

$$\frac{\sin x}{\sin(x+y)} = \prod_{k=0}^{N-1} \frac{\sin\big((x+\pi k)/N\big)}{\sin\big((x+\pi k)/N + y/N\big)},$$

$$\frac{\sin y}{\sin(x+y)} = \prod_{k=0}^{N-1} \frac{\sin\big((y+\pi k)/N\big)}{\sin\big(x/N + (y+\pi k)/N\big)},$$

$$\frac{\sin x}{\sin y} = \prod_{k=0}^{N-1} \frac{\sin\big((x+\pi k)/N\big)}{\sin\big((y+\pi k)/N\big)}.$$

By (2.1) the left sides are $\sin(x,y)$, $\cos(x,y)$ and $\tan(x,y)$, and each factor is the general function at the stated arguments. No denominator vanishes, because the product of the denominators is $\sin(x+y)/2^{N-1} \neq 0$, or $\sin y/2^{N-1} \neq 0$ for the tangent. □

For $N = 2$ the first identity reads $\sin(x,y) = \sin(\frac{x}{2}, \frac{y}{2})\, \sin(\frac{x}{2} + \frac{\pi}{2}, \frac{y}{2})$, and repeating it n times gives the case $N = 2^n$, the form in which the identity appears in [2].

As N grows, the parts x/N, y/N and $(x+y)/N$ of the arguments become small while the shifts $\pi k/N$ stay exact. Write $u_{N,k} = \sin(\pi k/N)$ and $v_{N,k} = \cos(\pi k/N)$. For small s, $\sin(s + \pi k/N) = v_{N,k} \sin s + u_{N,k} \cos s \approx v_{N,k}\, s + u_{N,k}(1 - s^2/2)$. With $s = z/N$ and after multiplication by $-2N^2$, each sine in the factors of Theorem 7.1 becomes the quadratic

$$P_{N,k}(z) = u_{N,k}\, z^2 - 2N\, v_{N,k}\, z - 2N^2 u_{N,k}. \tag{7.5}$$

Definition 7.2 (General Sinus Transform)**.** The *General Sinus Transform of order N* is the triple of rational functions

$$S_N(x,y) = \prod_{k=0}^{N-1} \frac{P_{N,k}(x)}{P_{N,k}(x+y)}, \qquad C_N(x,y) = \prod_{k=0}^{N-1} \frac{P_{N,k}(y)}{P_{N,k}(x+y)},$$

$$T_N(x,y) = \prod_{k=0}^{N-1} \frac{P_{N,k}(x)}{P_{N,k}(y)}.$$

Each factor is a ratio of two quadratic polynomials, and S_N, C_N, T_N approximate $\sin(x,y)$, $\cos(x,y)$, $\tan(x,y)$ with products in place of the sums of a truncated Taylor series. The approximation is not exact for any finite N, since each factor carries the error of the second-order expansion of $\sin s$ and $\cos s$. Surprisingly, the approximation itself can be computed in closed form, because the truncated factors are again sines, of slightly enlarged angles.

For $|\varepsilon| < \sqrt{2}$ let

$$\rho(\varepsilon) = \sqrt{1+\varepsilon^4/4}, \qquad \delta(\varepsilon) = \arctan\frac{\varepsilon}{1-\varepsilon^2/2}, \tag{7.6}$$

so that $1-\varepsilon^2/2 = \rho\cos\delta$ and $\varepsilon = \rho\sin\delta$, and $\delta(\varepsilon) = \varepsilon + \varepsilon^3/6 + O(\varepsilon^5)$.

Theorem 7.3 (Closed form of the transform)**.** *Let $x, y \in \mathbb{R}$ and let $N > \max(|x|, |y|, |x+y|)/\sqrt{2}$. Then, whenever the denominators below do not vanish,*

$$\begin{aligned} S_N(x,y) &= \Big(\frac{\rho(x/N)}{\rho((x+y)/N)}\Big)^N \frac{\sin\big(N\delta(x/N)\big)}{\sin\big(N\delta((x+y)/N)\big)}, \\ C_N(x,y) &= \Big(\frac{\rho(y/N)}{\rho((x+y)/N)}\Big)^N \frac{\sin\big(N\delta(y/N)\big)}{\sin\big(N\delta((x+y)/N)\big)}, \\ T_N(x,y) &= \Big(\frac{\rho(x/N)}{\rho(y/N)}\Big)^N \frac{\sin\big(N\delta(x/N)\big)}{\sin\big(N\delta(y/N)\big)}. \end{aligned} \tag{7.7}$$

Proof. Let $\varepsilon = z/N$ and $\vartheta_k = \pi k/N$. By (7.5),

$$\begin{aligned} P_{N,k}(z) &= -2N^2\Big(u_{N,k}\big(1-\tfrac{\varepsilon^2}{2}\big) + v_{N,k}\,\varepsilon\Big) = -2N^2\rho(\varepsilon)\big(\sin\vartheta_k\cos\delta + \cos\vartheta_k\sin\delta\big) \\ &= -2N^2\rho(\varepsilon)\,\sin\big(\vartheta_k + \delta(\varepsilon)\big). \end{aligned}$$

By (7.1), $\prod_{k=0}^{N-1} P_{N,k}(z) = (-2N^2\rho(\varepsilon))^N \sin(N\delta(\varepsilon))/2^{N-1}$. Taking the quotients that define S_N, C_N and T_N, the factors $(-2N^2)^N/2^{N-1}$ cancel. □

The theorem says that the transform evaluates the general functions at slightly perturbed angles: $S_N(x,y)$ is, up to the factor $(\rho(x/N)/\rho((x+y)/N))^N = 1 + O(N^{-3})$, the general sine at $\big(N\delta(x/N),\, N\delta((x+y)/N) - N\delta(x/N)\big)$, and $N\delta(z/N) = z + z^3/(6N^2) + O(N^{-4})$. This gives the rate of convergence exactly.

Corollary 7.4 (Rate of convergence)**.** *For fixed x, y with $x + y \notin \pi\mathbb{Z}$, as $N \to \infty$,*

$$S_N(x,y) = \sin(x,y) + \frac{c(x,y)}{N^2} + O\big(N^{-3}\big),$$

where

$$c(x,y) = \frac{x^3\cos x\,\sin(x+y) - (x+y)^3\sin x\,\cos(x+y)}{6\sin^2(x+y)}.$$

In particular $S_N \to \sin(x,y)$, and similarly $C_N \to \cos(x,y)$ and $T_N \to \tan(x,y)$, each with an error of exact order N^{-2} unless the coefficient vanishes.

Proof. Insert $N\delta(z/N) = z + z^3/(6N^2) + O(N^{-4})$ and $\rho(z/N)^N = (1 + z^4/(4N^4))^{N/2} = 1 + O(N^{-3})$ into (7.7), and expand $\sin(z+h) = \sin z + h\cos z + O(h^2)$ in numerator and denominator. □

Remark 7.5 (Numerical behaviour). At $(x,y) = (0.3, 2.7)$, close to the pole $x + y = \pi$, the coefficient is $c = 66.14$, and the computed values of $N^2(S_N - \sin(x,y))$ are 66.28 for $N = 64$ and 66.12 for $N = 1000$. Over the four points $(0.5, 0.7)$, $(1,1)$, $(2, 0.9)$ and $(0.3, 2.7)$, the largest error of S_N is about 1.6×10^{-2} for $N = 64$, 6.3×10^{-5} for $N = 1024$ and 2.5×10^{-7} for $N = 16384$. The coefficient $c(x,y)$ grows like $\sin^{-2}(x+y)$ near the poles, where $\sin(x,y)$ itself is unbounded. In computations the values $u_{N,k}$, $v_{N,k}$ must be accurate: generating them by a first-order rotation recurrence adds an error that accumulates over the N factors and dominates the error of the transform.

The transform is of structural rather than computational interest: evaluating S_N costs $O(N)$ operations, far more than computing $\sin x/\sin(x+y)$ directly. What it shows is that each general function is, exactly, a product of copies of itself at small arguments (Theorem 7.1), and that each copy is close to a ratio of quadratic polynomials, much as the product formula $\sin x = x\prod_{m\geq 1}\big(1 - x^2/(m\pi)^2\big)$ expresses the classical sine.

8. Verification

The script `verify_gsp.py`, which accompanies the paper, checks every numbered statement. Identities are evaluated at 50 significant digits at random points of their domains, with independent random draws for each identity; derivatives, including D^n for $n = 1, \dots, 6$ and the primitives of Corollary 6.5, are computed symbolically with SymPy and compared at 50 digits; the product identities are tested for every N from 1 to 40; the closed form of Theorem 7.3 is compared with the products themselves and the coefficient of Corollary 7.4 with the computed errors; the fixed-frame formulas of Proposition 6.2 are tested for n from -3 to 5; and the numbers quoted in Remark 7.5 are recomputed.

9. Conclusion

Projecting a point of the circle onto a line along an arbitrary direction, instead of straight down, gives an oblique triangle, and the ratios of its sides are the general sine, cosine and tangent. They are the coordinates of the unit circle in an oblique frame (Proposition 3.1) and satisfy the oblique Pythagorean identity (3.2). Their addition formulas, in the studied and in the projection angle, all reduce to Ptolemy's identity (Theorems 5.2 and 5.3). With the frame held fixed, they obey addition formulas with a single cross term and De Moivre's formula (Proposition 6.2), and differentiation turns the point by a quarter turn, so that $D^2 f = -f$ (Theorem 6.4). They are exact products of their own copies at scaled arguments, for any number of factors (Theorem 7.1); truncating each factor gives a rational approximation which is itself a ratio of sines in closed form, and whose error is of exact order N^{-2} (Theorem 7.3 and Corollary 7.4).

Two questions remain open. The first is whether the general functions satisfy orthogonality relations, for a suitable weight, that parallel those of the classical functions; with the frame fixed, C_θ and S_θ are combinations of $\cos\theta$ and $\sin\theta$, but the natural inner product of the oblique frame is not the orthonormal one. The second is the analogue of the construction for the hyperbolic functions, where the oblique frame of the Euclidean plane is replaced by a frame of the split-complex plane [9, 5].

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